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Concepts, Variables, and Measurement

Introduction

What we consider to be real or not real is often a matter of personal experience and social standing. For example, one person might argue that racism does not exist in the United States because laws protect against racial discrimination. Another person might argue that racism does exist because he or she has experienced it firsthand, growing up in a community of color. Which claim is correct? While it is true that many laws do protect against

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Life chances A phrase that Max Weber used in his analysis of social class to refer to differences in the likelihood of people from different class backgrounds having access to similar resources. Differential access to resources can lead to different opportunities (chances) in life.

acts of racial discrimination, it is also true that many communities of color have underfunded schools that limit their students' *life chances.*

In fact, 2000 Census data from the state of New York show that per capita income among whites is $27,244, and among blacks it is only $15,498. The difference reflects historical and institutionalized forms of discrimination (access to loans, education, jobs, and other pathways to upward mobility), not individual deficiencies. Thus while one person may be correct stating that racism does not exist because laws protect against it, another person might also be correct in stating that racism does exist due to structural barriers to personal success.

Unlike a rock analyzed by geologists, it is impossible to go outside, pick up some racism, and bring it back inside to analyze. Yet we know it exists by looking for examples in our own lives or in academic literature. In his book *Sundown Towns* (2005), sociologist James Loewen documents the existence of hundreds of towns across the United States that banned blacks from being within the town limits between sunset and sunrise. Similarly, sociologist Robert Bullard (1994) documents the tendency for communities of color to bear a disproportionate burden of ecological hazards relative to white communities.

The question may not be *Does racism exist?* Instead, the question may be *Depending on how you define it, does racism exist?* It is therefore extremely important to be very careful in how we define concepts, particularly if we are going to use those concepts to collect and statistically analyze data from the world around us.

Table 1 contains some important terms and definitions that are used in this chapter.

This chapter focuses on some of the most important underlying ideas behind statistics. It lays a foundation upon which we can move forward into actual statistical analyses. It is divided into three main sections: Concepts, Variables, and Measurement. Concepts are ideas that we hope to be able to effectively operationalize into variables. Variables are characteristics of people, or other units of analysis, that vary from case to case. Measurement refers to the considerations that we must take into account when creating variables.

TABLE 1

TERM	DEFINITION
Variable	Anything that varies from case to case *or* a logical set of attributes
Attribute	A characteristic of a variable
Conceptualization	The process of defining what is meant by a variable
Operationalization	The process of creating an actual measure for a variable
Level of measurement	Levels of measurement include nominal, ordinal, and interval/ratio
Nominal variable	A variable for which the attributes cannot be ranked from high to low
Ordinal variable	A variable for which the attributes can be ranked from high to low, but do not take numeric form
Interval/ratio variable	A variable for which the attributes can be ranked from high to low because they take numeric form
Mutually exclusive	A condition that exists when a case fits into one and only one attribute of a variable
Collectively exhaustive	A condition that exists when attributes of a variable exhaust all the possible values that cases may have
Validity	The degree to which a measure reflects the idea it was intended to measure
Reliability	The degree to which a measure yields consistent results across samples

Concepts

Concept An idea that we think represents something in the real world.

The Earth is Round and not the Center of the Universe? Imagine yourself in a world where many people believe that the world is flat and that if ships sail too far toward the horizon, they will drop off the edge of the Earth. Other people believe that the Earth is round and located at the center of the universe because the stars, Moon, and Sun appear to circle the Earth. Without evidence, how do we determine who is right?

One day, a group of sailors who had left years earlier on an expedition return, claiming that they have sailed *around* the world (Figure 1). Of course, if one believes that the Earth is flat, this claim is preposterous. For others, it reinforces the belief that the stars, Moon, and Sun circle the Earth. It also raises the possibility that the Earth is spinning and that it may not be the center of the universe. The movement of the stars may only give the illusion of the Earth being the center of the universe.These are dangerous claims that challenge fundamental belief systems and some people were persecuted and killed for such beliefs.

Over the years, evidence builds to support the claim that by sailing continuously in one direction, for example, west, a ship can actually end up where it began. If the only way that a line can end up where it began is by going around, then the ships must have sailed around the world. This finding presents a serious challenge to the dominant

Figure 1 **Magellan's Route Around the World**

understanding of the world and eventually leads to new ideas regarding gravity, the laws of motion, astronomy, and even religion.

Why is it that ships sailing toward the horizon do not lift off the round Earth and sail off into space? Something must keep them "stuck" to the Earth—but what? Gravity. Gravity must keep ships attached to the round Earth and prevents them from leaving the Earth's surface and sailing off into space. The idea of gravity existed for about 1,000 years before Sir Isaac Newton published *Principia*, but he is generally credited with developing some of the models that many scientists still use today.

Has anyone ever seen gravity? Has anyone ever touched gravity? No, of course not. Gravity exists at the conceptual level. Today, we believe that all objects are somehow attracted to one another because of gravity. We use these ideas to predict the motion of planets, the trajectory of comets, and the presence of black holes. It is an idea that works well and can explain a lot of physics. Yet, if we could jump forward in time, we might be surprised to find that another concept has come to replace our current concept of gravity.

In this example, gravity is a human construct that was a long time in the making; it is a concept or an idea that we use to help us make sense of the way the world works. Concepts are ideas that we think represent reality.Like our understanding of the earth and gravity, social science is based on the creation and analysis of concepts. Racism, sexism, inequality, love, and hate are all examples of concepts. They do not exist in the sense that we can pick them up, weigh them, or put them under a microscope; however, we know that they exist because we live them every day. Such is the stuff of sociological analysis.

Scientific Revolutions

In his groundbreaking book *The Structure of Scientific Revolutions* (1962), philosopher Thomas S. Kuhn referred to the shift from believing that the world is flat to believing the world is round as a scientific revolution. Scientific revolutions occur when old understandings of the way the world works are replaced with new explanations. Kuhn argues that people tend to think of science as a slow accumulation of knowledge over time. This gradual accumulation of knowledge is incorporated into increasingly sophisticated models of how the world works that are then used to pursue a more informed knowledge of reality. Many people believe that this scientific "progress" is the way to find the truth that lies behind our misunderstandings of the universe.

Kuhn argues that the search for "truth" is misguided. He and many others argue that we never find the truth, but we do develop better and more sophisticated understandings of the world around us. For example, when the "flat earth" theory is challenged to the point of not working, a scientific revolution takes place in which old interpretations of the world are replaced with new. Given our relatively limited investigations into these topics, why should we expect that our new model will be the final one?

Michael Interisano/Design Pics/Newscom

While the world around us exists physically, and many of our ideas about how it works seem to work quite well, our understanding of it is based on imperfect models and, often, flawed interpretations. In other words, the concepts developed to make sense of our surroundings often have shortcomings. Nevertheless, scientists use the best models available to them at the time until a new model is developed that does a better job. We refer to scientific eras when a single model of the world dominates other possible understandings as times of "normal science." The problem with these times is that anomalies tend to arise. Anomalies are scientific findings that cannot be explained by the dominant model(s) of the time.

Anomalies Phenomena that are not explained by existing models of understanding.

Take the example of learning that the world is not flat. A ship sailing around the world is an anomaly because it never should have happened if the world is flat. The ships should have fallen off the edge of the world, not returned to their point of origin. The events do not fit the current model of the Earth and they challenge us to find a new model with which to understand the world, and reconsider our current astronomy.

A couple of anomalies, however, do not always necessitate the overhaul of a system of logic and beliefs. It may be that those proclaiming the anomaly are persecuted for their beliefs, or that the anomaly itself was a misinterpretation of events. How many anomalies it takes to necessitate a change in the current paradigm is often a question of politics and power more than it is a question of legitimacy (particularly when religious beliefs are challenged). Let's just say that when enough evidence is accumulated to offer support for the claim that the world is not flat, a scientific revolution takes place in which old understandings are jettisoned for new models that work "better." On the following page is a box titled *Now You Try It*. These boxes are located throughout each chapter and are intended to provide readers with opportunities to check their understanding of the concepts as they are presented. The answers to the *Now You Try It* exercises are located at the end of each chapter. We now turn our attention to a sociological concept that seems to be withstanding the test of time.

Now You Try It

Here is an exercise that may help you realize just how much we take the world for granted. Have you ever stopped to wonder about the location of the moon in the night sky? The answer to the question posed below can be had in a couple of ways: quickly (by looking at the end of this chapter) or slowly (by gathering data over the next several days).

Most people know where the sun rises and where it sets. The sun rises in the east and sets in the west. But most people do not know where the moon rises and sets; or where Saturn rises and sets; or any heavenly body for that matter. So where does the moon rise and where does it set? Can you offer a logical explanation for your answer?

Answer at the end of the chapter.

Emile Durkheim, Structural Strain, Suicide, and Anomie

In 1912 Emile Durkheim (1858–1917) became the first person to ever hold the official title of "sociologist" at a university (at the Sorbonne in Paris). Although he studied and wrote on a number of sociological topics, including the division of labor, religion, education, and social control, he is probably most well known for his studies of suicide. An astute statistician, Durkheim poured over thousands of government documents to try and make sense of suicide, a normal part of all societies. What intrigued him, however, was that suicide rates change over time. Ultimately, he concluded that in times of great uncertainty, or what we might think of as times of normlessness, suicide rates go up. He called this state of normlessness *anomie*.

Anomie A societal condition in which the normative standards of behavior are unclear.

Durkheim's concept of anomie is one example of a sociological concept that has withstood the test of time, a concept that he used to help better understand changes in suicide rates. And it appears that Durkheim's concept may be just as applicable today as it was in the early part of the 20th century when he developed it.

Sociologists have studied suicide for about a century. We know that the suicide rate in the United States is about 11, meaning that in any given year 11 people out of every 100,000 people will take their own lives. Another 600,000 will be admitted to emergency rooms for self-inflicted injuries (Centers for Disease Control, 2009). Of these roughly 33,000 people, males between the ages of 16 and 26 and males over the age of 70 are two of the demographic groups most likely to commit suicide. Why males? Why males of these age ranges? Explaining why requires the use of concepts.

As you can see from Table 2, the suicide rate is significantly higher for men than it is for women, a trend that holds true in every state. This raises interesting questions regarding differences in how males and females think about suicide and how they act on those ideas.

Durkheim is often noted for his contributions to our understanding of the conditions in which suicide is more likely to occur. He referred to certain characteristics of groups or communities as "social facts." In other words, he believed that social forces exist externally from individuals and these forces "push" and "pull" our actions into predictable patterns, including patterns pertaining to rates of suicide.

TABLE 2 *USA State Suicide Rates and Rankings by Gender, 2005*

2005	NATION - BOTH SEXES COMBINED		2005	MEN		2005	WOMEN	
Rank	State	Crude Rate	Rank	State	Crude Rate	Rank	State	Crude Rate
1	Montana	22.0	1	Montana	36.2	1	Wyoming	9.2
2	Nevada	19.9	2	Nevada	30.9	2	Nevada	8.5
3	Alaska	19.8	3	Alaska	30.4	3	Alaska	8.4
4	New Mexico	17.8	4	New Mexico	28.9	4	Montana	7.9
5	Wyoming	17.7	5	Colorado	27.0	5	Colorado	7.2
6	Colorado	17.2	5	North Dakota	27.0	6	New Mexico	6.9
7	Idaho	16.0	7	South Dakota	26.9	7	Arizona	6.7
8	Arizona	15.9	8	Idaho	26.8	8	Oklahoma	6.6
9	South Dakota	15.6	9	Wyoming	26.0	9	Oregon	6.4
10	Oregon	15.4	10	Arizona	25.0	9	Florida	6.4
11	Oklahoma	14.7	11	Oregon	24.5	11	Arkansas	6.1
12	North Dakota	14.5	12	Tennessee	24.0	12	Vermont	5.7
13	Arkansas	14.4	13	West Virginia	23.8	13	North Carolina	5.5
13	Tennessee	14.4	14	Oklahoma	23.1	14	Kentucky	5.3
15	West Virginia	14.1	14	Arkansas	23.1	14	Washington	5.3
16	Utah	14.0	16	Utah	22.9	16	Tennessee	5.2
17	Kentucky	13.6	17	Maine	22.5	16	Iowa	5.2
18	Maine	13.3	18	Kentucky	22.2	16	Hawaii	5.2
19	Florida	13.2	19	Kansas	22.0	19	Utah	5.0
19	Kansas	13.2	20	Mississippi	21.1	20	Idaho	4.9
21	Washington	13.1	21	Washington	20.9	20	South Carolina	4.9
22	Missouri	12.5	22	Missouri	20.8	22	Virginia	4.8
22	Vermont	12.5	23	New Hampshire	20.7	22	West Virginia	4.8
22	Mississippi	12.5	24	Florida	20.3	24	Missouri	4.7
25	New Hampshire	12.4	25	Indiana	20.2	25	Nebraska	4.6
26	South Carolina	12.0	26	Alabama	19.9	25	Louisiana	4.6
27	Indiana	11.9	27	Ohio	19.7		Total	4.5
28	Alabama	11.8	28	Vermont	19.6	27	Michigan	4.5
29	Ohio	11.7	29	South Carolina	19.5	27	Kansas	4.5
30	North Carolina	11.6	30	Pennsylvania	19.2	27	Maine	4.5
30	Wisconsin	11.6	31	Wisconsin	19.0	30	Mississippi	4.4
32	Pennsylvania	11.5	32	Virginia	18.3	30	New Hampshire	4.4
32	Virginia	11.5	33	Louisiana	18.2	30	South Dakota	4.4
34	Iowa	11.2	34	North Carolina	18.1	30	Wisconsin	4.4

(Continued)

2005	NATION - BOTH SEXES COMBINED		2005	MEN		2005	WOMEN	
Rank	State	Crude Rate	Rank	State	Crude Rate	Rank	State	Crude Rate
34	Louisiana	11.2		Total	17.8	34	Texas	4.3
	Total	11.0	35	Michigan	17.6	34	Pennsylvania	4.3
36	Michigan	11.0	36	Iowa	17.4	36	Alabama	4.1
37	Minnesota	10.7	36	Minnesota	17.4	36	California	4.1
38	Nebraska	10.6	38	Delaware	16.9	36	Ohio	4.1
38	Texas	10.6	38	Texas	16.9	39	Minnesota	4.0
40	Georgia	10.1	40	Nebraska	16.8	40	Georgia	3.9
41	Delaware	9.9	41	Georgia	16.6	41	Indiana	3.8
42	California	8.9	42	Maryland	14.1	41	Rhode Island	3.8
43	Illinois	8.5	43	Illinois	13.8	43	Massachusetts	3.6
44	Maryland	8.4	44	California	13.7	43	Connecticut	3.6
44	Connecticut	8.4	45	Connecticut	13.6	45	Illinois	3.4
44	Hawaii	8.4	46	Hawaii	11.6	46	Delaware	3.2
47	Massachusetts	7.5	46	Massachusetts	11.6	46	Maryland	3.2
48	Rhode Island	6.6	48	New York	10.4	48	District of Columbia	2.9
49	New Jersey	6.2	49	New Jersey	9.9	49	New Jersey	2.6
49	New York	6.2	50	Rhode Island	9.6	50	New York	2.2
51	District of Columbia	5.7	51	District of Columbia	8.8	51	North Dakota	1.9

Data Source: CDC's WISQARS website "Fatal Injury Reports," http://www.cdc.gov/ncipc/wisqars/; downloaded 24 January 2008 "Crude rate" refers to rates per 100,000 population. Prepared by John L. McIntosh, Ph.D., Indiana University South Bend for posting by the *American Association of Suicidology* (www.suicidology.org) - January 2008.

Suicide exists in all societies (to varying degrees), but the number of suicides per 100,000 people (the suicide rate) is likely to vary depending on social conditions. To test this idea, Durkheim analyzed historical data on the number of suicides from one year to the next. Among other trends, he found that males were more likely to commit suicide than females and that Protestants were more likely to commit suicide than Catholics. Why?

Durkheim was fortunate to be a contemporary of other well-known sociologists, namely Karl Marx (1818–1883) and Max Weber (1864–1920). This allowed him to combine his own ideas with those of his contemporaries, and in particular, ideas about the workings of capitalism and the impact that capitalism may have on people.

Durkheim theorized that all people are inherently social beings and that humans need one another to (1) survive materially and (2) set limits on our worldly desires. He argues that without a society to set normative standards for the accumulation of wealth and other worldly possessions, the capacity for humans to constantly want "more" is endless. This is important because, as Max Weber showed in his famous book *The Protestant Ethic and the Spirit of Capitalism*, the confluence of the Protestant ethic (hard work, frugal living, and reinvestment as a way to accumulate wealth to demonstrate one's worth to God) and the spirit of capitalism (an ever-expanding economy with seemingly limitless capacity for growth) created the very conditions in which vast quantities of material wealth can be generated.

Anomie Then. Analyzing how suicide rates vary over time and from one society to the next, Durkheim (1897/1951) concluded that people were more likely to commit suicide during times of uncertainty, for example, during economic depression. Out of this finding he developed his concept of anomie. Anomie is a condition in society that increases when the established rules and regulations of society become blurred or no longer seem to apply. An example of this is a questioning of the belief in free market economies during times of economic crisis.

When rules become less clear, or in the absence of rules, people are more likely to suffer as their society is increasingly characterized by a state of pathological anomie (unhealthy or extreme levels of anomie). It is during these times that Durkheim detected increased rates of suicide.

At particular risk of suicide during times of economic slowdown are males and Protestants. Higher societal expectations for economic success among males can result in greater levels of anomie among men during economic recessions because males are more likely to experience declining social status. Similarly, because the accumulation of wealth has greater religious significance to Protestants than to Catholics (according to Weber), economic downturns can create the belief of less favorable outcomes in the afterlife (going to heaven) and generate greater levels of anomie. Therefore, we can predict and explain higher suicide rates among males and Protestants during times of economic decline. And Durkheim's concept of anomie remains particularly useful for explaining the high rate of suicide among young men today.

Anomie Now. A recent article in *Newsweek* magazine based on the work of sociologist Michael Kimmel (who recently published a book titled *Guyland* [2008]) notes that American males between the ages of 16 and 26 have one of the highest rates of suicide in the country. It is interesting that this rate seems to coincide in an era of extended juvenile behavior for this group. Twenty-something men in the late 1990s and early 2000s are much more likely to be extending their college party lifestyle for an additional decade or more than are any of their historical counterparts. The rise in suicide rates and the delay of life course changes may both be explained sociologically.

A 30-plus year contraction in wages, limited opportunities for upward mobility, and declining labor force status relative to females have created the conditions in which male college graduates face one of the toughest job markets in history. It may be that as the relative social status of males declines young males entering the workforce are more likely to be confronted with the question of what it means to be male. Thus the social context in which anomie rises begins to emerge. As the societal norms of "maleness" are increasingly blurred by the economic downturn, suicide rates are likely to increase.

Some sociologists have noted that mass media corporations are capitalizing on the newfound status of young males by promoting a culture that endorses partying, lack of responsibility, and objectification of women (Kimmel, 2008). These young males appear

SHANNON STAPLETON/Reuters/Landov

to be victims of the strain that results from high societal goals for achievement (high income, material wealth) without the means (strong job markets, expanding economy) to reach them. Consequently, they are more likely to turn to media portrayals of what it means to be male as cues for their own behavior and worldviews.

To summarize, Durkheim, like all scientists (chemists, physicists, biologists, botanists, psychologists, sociologists, and others), looked for patterns. Upon identifying a pattern, the goal is to explain it. To explain changes in suicide rates, Durkheim developed the concept of anomie.

New models of the way the world works enable us to make sense of our surroundings, but these models are often based on highly abstract ideas like anomie. While this is not necessarily a problem, it does require scientists to remain relatively open-minded about the possibility of flaws in their models. Identifying and explaining patterns among variables is a difficult and often abstract process.

Why Statistics?

How does statistics contribute to our understanding of the world? Why should students take an entire course on statistics? These are good questions for which there are some good and some not-so-good answers. The not-so-good answers tend to go something like this: “It is required for the degree.” “I had to take it, so should you.” “You can’t be a good social scientist without statistics.” Actually, you can be a very good social scientist without statistics, but you can be even better with some basic statistical tools at your disposal. This brings us to the good answers to the question, *Why statistics?*

Statistics Numeric representations of reality that facilitate our ability to describe, communicate, predict, and act.

For better or worse, we live in a world of statistics, statistical reasoning, and decisions made on the basis of statistics. In the most basic sense, statistics are numerical representations of reality. Often they are valid representations, other times they are not so valid. Statistics are used to describe conditions (such as average income), communicate more effectively (baseball fans understand the meaning of a .300 hitter), predict outcomes (such as the likelihood of a baseball player getting a hit), and develop policy (such as whether a drop of 3,000 points in the Dow Jones Industrial warrants a financial bailout package). In this sense, statistical knowledge at the individual level provides deeper insight into the kind(s) of forces that make the world what it is.

A. Ramey/PhotoEdit, Inc.

In his widely acclaimed book *Damned Lies and Statistics: Untangling Numbers from the Media, Politicians, and Activists* (2001), sociologist Joel Best argues that there are several kinds of statistics “consumers” in the world: the Awestruck, the Naive, the Cynical, and the Critical. Best defines the Awestruck as those who fail to think critically about statistics and are likely to believe what they hear about a statistic (p. 162). The Naive consist of those who are “basically accepting” of statistics because they tend to feel that most people, including statisticians, are sincere (pp. 162–163). The Cynical are those who tend to reject statistics on the basis that they amount to nothing more than sophisticated lies that are generated out of particular interests (p.164). Finally, Best discusses the Critical consumers. These are people who understand that every statistic is a complex way of conveying information. They tend to know something about statistics and they tend to know what kinds of questions to ask to clarify confusion or misleading statements (pp. 166–167).

The purpose of taking a statistics course is not to learn how to calculate a bunch of different statistics, although you will be asked to do that. Instead, the purpose of such a course is to increase the range of analytical tools that we have to approach and live in the world with which we are confronted. It is to learn to think critically and reflectively about what others tell us about the world so that we can come to our own, well-informed conclusions. In this sense, statistical literacy is really no different from other types of literacy (language, computer, financial) or skills (driving, cooking, home repair) that we need to navigate our daily routines.

The goal of this text is to provide you with an introduction to some of the most important statistical concepts (central tendency, dispersion, probability, and association). Real-world examples and data are used as much as possible to try and convey the manners in which these tools and skills can be applied. Examples with step-by-step calculations are included. Finally, the book is based on the belief that by working through a few calculations, students are more likely to grasp the concepts and be able to apply them more effectively.

Variables

Conceptual Definitions of Variables

Scientific analysis tends to consist of making sense of (1) describing how cases are distributed and (2) describing how variables interact with one another. For the most part, very few constants exist in science and really none exist in the social sciences. Everything varies from case to case. As we move from one person to the next, sex, age, hair color, education, and food preferences all vary. This is true when we compare individuals; and it is true when we compare families, communities, cultures, and any other *unit of analysis*. Scientists in general, but particularly social scientists, work with what are called *variables*. Two common ways of thinking about variables are (1) anything that varies from case to case and (2) logical groupings of *attributes*.

> *Units of analysis* are scientists' objects of analysis, or who or what is being studied.
>
> A variable is defined here as anything that varies from one case to the next. For example, the sex of a respondent in a survey might vary between males and females. Variables are the building blocks of scientific research.
>
> *Attributes* are defined here as a set of logical characteristics for a variable. Logical groupings of attributes constitute variables. For example, the attributes *male* and *female* constitute the variable Sex of Respondent.

Units of analysis The who or what that data describe. Tend to consist of individuals or groups/places (towns, schools, teams, etc.).

Variable Anything that may vary from one case to the next (sex, height, level of education, etc.).

Attributes A logical set of characteristics for a variable.

In social science, variables can measure ideas or physical attributes. Physical attributes are easy to comprehend. For example, we might want to describe the members of a college sports team. The players' height, weight, age, and sex are all easily described variables. We might then want to describe their overall academic success. To do this we might check their SAT scores, their overall GPA, or whether they have made the Dean's List. But do any of these really give us a true reflection of their level of academic success? Maybe, but some ideas are harder to measure than others.

HARTMANN CHRISTIA/SIPA/Newscom

For example, take the concept of *hate*. Hate exists only at the conceptual level. It is not a physical reality, though it has very real consequences. We might see examples of actions that we feel are motivated by *hate*, such as gay-bashing, shouting racial slurs, or genocide; but we still cannot pick up the hate, put it on a scale, and measure it. We must first define exactly what we mean by *hate* before we can find ways to measure it. After we have measured it in some methodologically valid and reliable manner, we can describe its characteristics statistically. The following excerpt is from the Federal Bureau of Investigation Uniform Crime Report and describes what that organization considers to be a hate crime.

> The FBI collects data regarding criminal offenses that are motivated, in whole or in part, by the offender's bias against a race, religion, sexual orientation, ethnicity/national origin, or disability and are committed against persons, property, or society. Because motivation is subjective, it is difficult to know with certainty whether a crime resulted from the offender's bias. Moreover, the presence of bias alone does not necessarily mean that a crime can be considered a hate crime. If law enforcement investigation reveals sufficient evidence to lead a reasonable and prudent person to conclude that the offender's *actions* were motivated, in whole or in part, by his or her bias, then the incident should be reported as a hate crime. (http://www.fbi.gov/ucr/hc2006/methodology.html, September 18, 2008)

Hypothesis A prediction about the distribution of a variable or the association between two variables.

Independent variable The variable that is controlled or held constant in a hypothesis.

Dependent variable The variable that is influenced by changes in the independent variable.

Direction of effect The part of a hypothesis that predicts whether two variables are positively or negatively associated.

As this shows, it is very difficult to determine a motivation and therefore it is very difficult to classify crimes as hate crimes. It is likely that racism, sexism, and other kinds of bias are responsible for a far greater number of crimes than are documented by the FBI.

In Durkheim's analysis of suicide, the unit of analysis is societies (which he defined very loosely) while anomie and suicide are variables (because they vary across time and place). The level of anomie in any given society changes over time. And the level of anomie tends to vary from one society to the next. Anomie, like hate, is not something that can be seen with the naked eye; yet it seems to be a reality because it makes sense logically and it is supported by historical data. "Seeing" anomie requires abstract thought and theoretical understanding. And as you may have already guessed, debates over how to measure concepts like anomie, that may or may not exist in reality, are common.

Independent and Dependent Variables

Statistical association is briefly introduced here. The idea behind statistical association is that we are working with two (or more) variables; one is called the independent variable and the other is called the dependent variable. We predict (or hypothesize) that a change in one variable, for example education, is associated with a change in another variable, income. When we say that education and income are associated, we are hypothesizing (making an educated guess or an informed prediction). To use more scientific language, we *hypothesize* that education and income are positively associated. The positive wording refers to the direction of effect. This means that we predict that increases in education are accompanied by increases in income. A negative direction of effect would exist when increases in education are accompanied by decreases in income.

In this example, education is the independent variable and income is the dependent variable. In other words, our assumption is that a respondent's income is, at least partially, a result of how much education they have. A good way to remember which variable is independent and which is dependent is to phrase the hypothesis in an "*if . . . then . . .* " format. For example, *if education increases, then income increases.* In this format, the independent variable always follows the *if* and the dependent variable always follows the *then.*

It is important to note that an association between two variables does not mean that changes in the independent variable *caused* changes in the dependent variable. Scientists use the phrase *Association does not imply causation* to emphasize this fact. Just because people with higher levels of education may have higher incomes, it does not mean that their education caused them to have higher income. To effectively argue that two variables are associated, three criteria must be met:

1. The cause must precede the effect. This means that we would have to show that education came before higher income as opposed to higher income preceding greater levels of education. This could be very difficult because people with higher incomes can afford greater levels of education.
2. Changes in the dependent variable must be the result of change in the independent variable and not some preceding or intervening variable. In other words, it is possible that higher levels of income could be the result of any number of factors, one of which may be education.
3. The association must be present "often enough." This means that we may find cases in which education and income are not associated; however, a few cases do not threaten the overall trend. The question then becomes *How often must the prediction hold true?* This is a tricky question and represents a point at which science often becomes political and the subject of debate.

Operational Definitions of Concepts

In thinking about concepts and variables, it is useful to think chronologically. Defining what we mean by an idea must precede devising ways to measure our ideas. Describing or defining an idea that we think represents something in the real world is a process called conceptualization. Devising ways to measure the ideas we have conceptualized is called operationalization.

Conceptualization The process of defining what we mean by a concept.

Operationalization The process of developing a variable that measures a concept.

When we take an idea, a concept, and define what exactly we mean by it, we are conceptualizing. For example, when we take the concept of income and conceptualize it, we might refer to the amount of money a person makes at a given job in a given year. Or we might refer to the amount of money they make at their job added to the amount of money they earn from investments. Either technique is a valid way to conceptualize income.

When we take an idea, a concept, and devise a measure by which to compare one case to the next, we are operationalizing a variable. For example, we could operationalize the concept of sex as consisting of two attributes: male and female. In this sense, the term variable could be defined as a logical set of attributes. As in the example above we might conceptualize income to refer to the amount of money a person makes at their job in a given year; but this leaves several possibilities open. We might refer to the actual dollar amount ($32,515), or we might refer to a range of income ($0–19,999; $20,000–29,999; $30,000–39,999; etc.). In other words, there is no right or wrong way to operationalize concepts into variables (it depends more on what you want to know).

> Conceptualization is the process of describing what we mean by an idea.
>
> Operationalization is the process of devising a specific measure for a variable that we have conceptualized.

Most social scientists probably believe that anything can be measured (alienation, class conflict, anomie), but only recently do they have access to the kinds of technological resources to gather, analyze, and communicate data that we now have at our disposal. Social scientists claim that anything can be measured, even concepts (ideas with no "real" existence). So why couldn't we measure something as abstract as Marx's concept of alienation or Durkheim's concept of anomie?

Alienation A condition in which people suffer from a "disconnection" between themselves and their work, and disconnection from one another.

Alienation. Karl Marx argued that capitalism is a mode of economic production based on conflict between two classes of people: capital (aka the bourgeoisie) and labor (aka the proletariat). Class membership is determined by one's relationship to the means of production. Capital owns the means of production and labor does not. At work, members of the labor class sell their ability to do work (their labor power) to capital in exchange for a wage rate. Marx claims that this has the effect of alienating people from their labor because individuals no longer control their labor nor do they control the products of their labor.

Logically, this makes sense; however, can we measure alienation in some way to offer empirical evidence that it actually exists? As in the case of Durkheim's *anomie*, Marx's *alienation* exists at the conceptual level. Can you think of ways to operationalize alienation?

You may not care very much about what you do for work as long as you make enough money. On the other hand, you might prefer to sacrifice some income for work that gives you a great deal of satisfaction, meaning work that is less alienating. From this we might surmise that someone who likes the type of work they do is less alienated from their labor than someone who does not like the type of work they do.

Although we cannot pick up alienation and put it on a scale to see how much of it a person experiences, we can ask people whether or not they are likely to stay at a job for the income, or to sacrifice some income for a job that they like. Those who indicate that they would be willing to take less pay but have a job that they like would be those experiencing greater levels of alienation. In this way, we have taken the concept of alienation and operationalized it as a variable.

> Would you be willing to take a pay cut to have a job that you enjoy more than your current job?
>
> Yes
>
> No

One problem that arises with this variable is that it has a class bias embedded within it. In other words, people who don't make enough money to make ends meet are much more likely to answer "No." Therefore, we could only compare the responses from those respondents of relatively similar class standing. We could argue that those who answer "Yes" are experiencing greater levels of alienation from their work than those who answer "No."

Several factors must be considered when operationalizing a variable. Two of these are that the variable must be (1) *collectively exhaustive* and (2) *mutually exclusive*.

Collectively exhaustive A necessary condition for variables that is met when the attributes of a variable include every possible response.

Collectively Exhaustive. We cannot say that a variable is operationalized correctly unless the attributes of that variable are collectively exhaustive. Attributes are collectively exhaustive when an attribute exists for every possible case.

EXAMPLE 1

Attitude Toward the Quality of City Parks

Suppose we are conducting a study of city parks and we want to find out how people of different racial backgrounds feel about the quality of parks in different neighborhoods. We might only have enough resources to sample 100 respondents. In our random sample we expect our sample to break down as follows:

TABLE 3 *Which of the Following Would You Classify Yourself As?*

White	50%
African American	35%
Native American	5%
Asian	5%
Latino	5%
Total	100%

It is entirely possible, however, that we have a respondent from Cape Verde who does not feel that any of our categories of race are applicable to him or her. How do we handle cases like this?

Statistically, 100 respondents could be an insufficient number of respondents upon which to claim that we have statistically significant results, particularly if we operationalize the variable in a way that allows some attributes to contain very small numbers of respondents, such as one Cape Verdean. Therefore, instead of having an attribute for each ethnicity, we might operationalize our variable as: White, African American, and Other. Our data might then look like Table 4.

This attribute of "Other" allows us to make the variable collectively exhaustive, leaving no one out. It does, however, have the effect of "glossing" over potentially important subtleties in the data by failing to differentiate Native Americans, Asians, Latinos, and Cape Verdeans.

If we are concerned about being able to generate statistically significant results to Native Americans, Asians, or Hispanics, we must then sample a larger number of respondents. Then we can change the attribute "Other" to "Native American, Asian, Latino, Cape Verdean, and Other."

TABLE 4 *Which of the Following Would You Classify Yourself As?*

White	50%
African American	35%
Other	15%
Total	100%

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Mutually Exclusive. We cannot say that a variable has been operationalized correctly unless the attributes of that variable are mutually exclusive. Attributes are mutually exclusive when the attributes do not overlap each other. For example, we might ask people to tell us what their annual pre-tax income is by placing an X on the appropriate line.

As you can see in Table 5, there is a problem with the variable on the left. What if you make $10,000, $40,000, or $60,000? Which line do you place an X on? The problem is that the variable to the left is not operationalized such that the attributes are mutually exclusive. The variable to the right is operationalized such that the attributes are mutually exclusive.

Mutually exclusive A necessary condition for variables that is met when each case can be applied to only one attribute of a variable.

Table 6 has two problems. Can you explain what is wrong with it? First, the attributes are not collectively exhaustive. It is possible, and likely, that in our sample we will find a respondent who makes more than $80,000 per year. There is no attribute in our variable for such a respondent to indicate this to us.

The second problem deals with mutual exclusivity. As you can see, a person making $10,000, $40,000, or $60,000 could respond to either of two attributes. It is vital that statisticians be aware of the sources of their data to avoid these kinds of problems. A common saying in statistics is "garbage in, garbage out." This means that if the data used to generate statistics are methodologically sound, then the statistics generated by that data are not representative of reality.

Remember that all variables must be operationalized such that the attributes are both collectively exhaustive and mutually exclusive. When variables do not meet these two requirements, data must be considered unreliable and invalid. Consequently, the statistics generated from such data must be considered suspect and nonrepresentative of the concepts they are intended to represent.

TABLE 5 *Pre-tax Income (2005)*

___ $0–10,000	___ Less than $10,000
___ $10,000–40,000	___ $10,000–39,999
___ $40,000–60,000	___ $40,000–59,999
___ $60,000 and higher	___ $60,000 and higher

TABLE 6 *Pre-tax Income (2005)*

___ $0–20,000

___ $20,000–40,000

___ $40,000–60,000

___ $60,000–80,000

Measurement

Levels of Measurement

Now that we have seen how social scientists are able to measure practically anything, provided the concept is operationalized in a valid and reliable manner, we can discuss *levels of measurement*. As with most tasks in life there is more than one way to skin a cat (whatever that means) and, as you might have guessed, there is more than one way to operationalize a variable.

Levels of measurement The degree of mathematical precision that can be applied to a variable. The three levels of measurement are referred to as nominal, ordinal, and interval/ratio.

For example, if I ask you if you like chocolate, you could answer *yes* or *no*. If I ask you how much you like chocolate, you could answer from the following list: *not at all*, *a little*, or *a lot*. Or I could have you indicate on a scale of 1 to 100 how much you like chocolate. A fundamental difference exists between these ways of measuring. One elicits responses that cannot be ranked from high to low and the others elicit responses that can be ranked from high to low. In the case of the 1–100 ranking, we can even count how far from 0 a person likes chocolate.

These are the three levels of measurement: nominal, ordinal, and interval/ratio. All variables can be categorized into one of these three levels, though sometimes the differences are not so clear. It is extremely important to understand the different levels of measurement and how to identify the level at which a variable is operationalized. The statistics that can be used to describe a variable depend on the level at which the variable is operationalized. In other words, if a variable is operationalized at the nominal level, only certain statistics can be used. For example, as we will see in the titled chapter "Frequency Tables Introduction", measures of central tendency (mean, median, and mode) are statistics used to describe where cases tend to cluster in a distribution. In the case of a nominal variable, only the mode can be used; however, in the case of an ordinal variable, both the mode and the median can be used. The following briefly describes each level of measurement and explains how they differ.

Nominal. Nominal variables have attributes that cannot be ranked from high to low. The attributes are nothing more than different categorical responses. For example, male or female are attributes of the variable of gender. Male is not more or less than female and female is not more or less than male. Similarly, we have the example of *religiosity*.

Nominal Variables operationalized at the nominal level have attributes that cannot be rank-ordered.

EXAMPLE 2

Religiosity

Suppose we ask 100 people whether Jesus (or any other religious figure) is part of their religiosity and get the results in Table 7.

TABLE 7 ***Is Jesus Part of Your Religiosity?***

Yes	50%
No	50%

TABLE 8 *How Strong Is Your Belief in Jesus?*

Not Strong	33%
Somewhat Strong	34%
Very Strong	33%
Total	100%

This does not mean that the people who answered "yes" are more religious. People who answer "yes" are not more or less religious than the people who answered "no"; it is just that some people have Jesus as part of their religiosity and some people do not. Therefore, the attributes cannot be ranked from high to low or more to less. They simply indicate a difference between two groups of people: those for whom Jesus is part of their religiosity and those for whom Jesus is not part of their religiosity.

Ordinal Variables operationalized at the ordinal level have attributes that can be rank-ordered, but those attributes do not reflect actual numeric values.

Interval/ratio Variables with attributes based on real or relative numeric values (meaning the attributes can be rank-ordered and used to conduct mathematical calculations).

Ordinal. Ordinal variables have attributes that can be ranked from high to low, but the "distances" between the attributes can't be measured. For example, consider a questionnaire that asks you how much you like a particular flavor of ice cream. Your choices are *a lot*, *somewhat*, and *a little*. A person indicating they like the flavor a lot obviously likes it more than someone who likes it a little. Yet we cannot measure exactly how much more the first person likes it.

In the religiosity example we see the same kind of logic. Suppose our data yield the results in Table 8.

Some peoples' belief in Jesus is "not strong," others is "somewhat strong," and others is "very strong." Therefore we know that a person who answers "very strong" has a stronger belief in Jesus than someone who answers "not strong." But how much stronger is it? We don't know because we have no way of measuring the "distance" between "not strong" and "very strong." We can rank-order our respondents from high to low in their belief in Jesus by placing them into one of the three attributes, but we have no way of addressing the question of how much stronger is one respondent's belief than the next.

Frank E. Lockwood/MCT/Newscom

Interval/Ratio. Interval/ratio variables have attributes that can be both ranked and measured. Attributes take the form of specific numbers, often with specified units of analysis. For example, the number of tackles that a football player has in a game varies from player to player. One player may have five tackles and another player has seven. The second one has more, *and* we know exactly how many more.

The same logic holds true for our measure of religiosity. When we ask respondents to tell us how many times they attend religious services in a month, they answer with a numeric amount. If one respondent attends religious services eight times a month and another attends one time each month, we know that the difference is seven. Respondents who attend more often are considered more religious, and therefore the first respondent might be considered to have eight times the religiosity as the second respondent. Not only do we know that the first respondent attends religious services more often, we know exactly how many more times.

Interval and ratio variables differ, but are treated the same statistically—that is why they are lumped into the same category. The difference between interval and ratio variables is that ratio variables have a true zero-point while interval variables have an arbitrary zero-point. For example, if I ask you how many skateboards you own and you say zero, then you own no skateboards. This is a ratio variable with a true zero. If I ask you how you rate the mayor on a scale of –10 to +10 and you say 4, then you have some approval of the mayor. Had you answered 0 you have still indicated some approval, but not as much as someone who answered 4.

The 0 on the scale of –10 to +10 is an arbitrary zero. The same is true for a temperature of 0 degrees. On the Fahrenheit scale, 0 degrees represents an arbitrary zero. It is an arbitrary point on a thermometer that happens to be 32 degrees below the temperature at which water freezes. On the Celsius scale, 0 degrees happens to be the temperature at which water freezes. This is an arbitrary point (why wouldn't we consider zero to be the point at which some other liquid freezes?). On neither the Fahrenheit nor the Celsius scale does 0 degrees represent the absence of heat.

Regardless of the level at which variables are operationalized, the attributes of all variables must be both **mutually exclusive** and **collectively exhaustive.**

Below is an example using a question that is often used for course and instructor evaluations. As this example shows, any concept can be operationalized in a number of ways. In the first technique, we have a yes or no question (a **nominal** variable). In the second, we incorporate the concept of degree (an **ordinal** variable). And in the third, we incorporate the concept of degree in a measurable manner (an **interval/ratio** variable). Each of these represents a different level of measurement.

EXAMPLE 3

Operationalizing Course Challenge

Variables can be conceptualized and operationalized in any number of ways.

Nominal: Did you find the course challenging? Yes No
Ordinal: How challenging was the course? Not at all Somewhat Very
Interval/Ratio: In a range from 0 to 100, how challenging was this course? ___

When gathering data, it is generally a good idea to collect the data at the highest level possible. For example, suppose you are doing research (a survey) on Grade Point Averages. You could ask respondents to circle which category their GPA falls within, as the box below shows. Or you could have respondents simply write down their GPA.

EXAMPLE 4

Surveying Student Grade Point Averages

TABLE 9 *GPA*

0.00–1.00
1.01–2.00
2.01–3.00
3.01–4.00

What is your GPA? _____

The advantage to having them write down their GPA, say it is 3.22, is that it allows you to see exactly what each respondent's GPA is. In other words, if you use the box above, you "gloss over" what could be important aspects of the data. For example, it could be that an important "cutoff" point in the data is 2.25, not 2.00; but by operationalizing it as in the box, this important trend might never be detected. Additionally, if you ask respondents to indicate their GPA, you can always go back to it and categorize it according to any grouping you desire.

Integrating Technology

Recoding Data

Statistical software programs allow researchers to *manipulate* data. Despite the negative connotation associated with the word *manipulate*, there is nothing ethically questionable about data manipulation. The term refers to the process of sorting cases, selecting particular cases, and other similar processes that allow for more detailed and effective statistical analysis of data. One common type of data manipulation is called "recoding."

When data are collected and entered into a computerized database, they contain numeric codes. For example, Male = 1 and Female = 2. In the case of nominal and ordinal variables the codes do not reflect actual value; they simply allow us to enter 1s and 2s into a database instead of typing out Male and Female. In the case of ratio (and often interval) variables, however, codes do reflect real values. For example, the number of years of education someone has might be 16. In this case, the number 16 is entered into the database.

Suppose we wanted to compare two different groups defined on the basis of whether or not they completed 12 years of education so that we could compare their income levels. By manipulating the data, we can quickly and easily create two groups of respondents – one group with less than 12 years of education and one with 12 or more years of education. We can assign a value of 1 to all respondents with less than 12 years of education and a value of 2 to all respondents with 12 or more years of education. In doing so, we essentially created a new variable by taking an interval/ratio variable and turning it into an ordinal variable. We know that every respondent in the second group has more years of education than any respondent in the first group, but we do not know the exact difference between any two respondents. This is *recoding*.

Data can only be recoded from higher-order levels of measurement to lower-order levels of measurement. In other words, we can recode interval/ratio variables into ordinal or nominal variables; and we can recode ordinal variables into ordinal variables with fewer attributes or into nominal variables. We cannot, however, recode in the other direction—nominal to ordinal or ordinal to interval/ratio. It is therefore important to try and collect data at the highest level of measurement possible because it provides a greater range of analytical possibilities later.

Take the example of attitude toward gun control. As Table 10 shows, gun control can be operationalized as a nominal, ordinal, or interval/ratio variable.

TABLE 10 *Gun Control as a Nominal, Ordinal, and Interval/Ratio Variable*

NOMINAL	ORDINAL	RATIO
Does your state have strict gun control laws?	How important of an issue is gun control to you?	On a scale of 0 to 10, how important is gun control to you?
Yes No	Very important Somewhat important Not important	Respondent indicates a number.

Graphical Representation of Data (What Data "Look" Like)

As we move from interval/ratio, to ordinal, to nominal levels of measurement, more and more information is lost and the range of statistical techniques that can be applied to the data diminishes. Therefore, it is usually a good idea to collect data at the highest level possible, for example, at the interval/ratio level, and then consider recoding it into ordinal or nominal later. This, of course, is a methodological consideration as much as it is a statistical consideration. While it is usually possible to recode downward (interval/ratio to ordinal to nominal), it is never possible to recode upward (nominal to ordinal to interval/ratio). The three charts below show how responses can be presented graphically.

Pie Charts. The chart in Figure 2 is called a pie chart. Pie charts are common ways of presenting data for nominal or ordinal variables. Remember that the attributes of nominal variables cannot be ranked from high to low. In this example, an argument could be made that if all the respondents were from the same state, the attributes could be ranked in some kind of high to low ordering. Often, what seem like clear-cut differences between nominal and ordinal variables turn out to have many more "shades of gray" than first appear.

Pie chart Often used with nominal and ordinal variables, pie charts consist of a circle cut into "pie slices" that add up to 100%. Each pie slice represents an attribute for the variable.

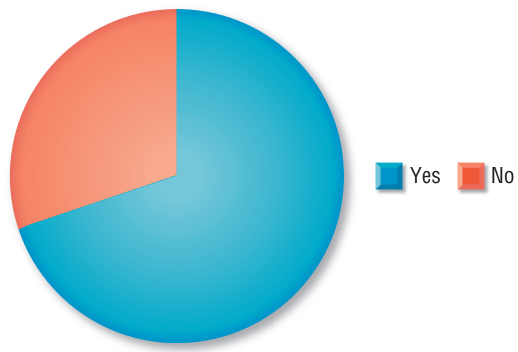


Figure 2 Does Your State Have Strict Gun Control Laws?

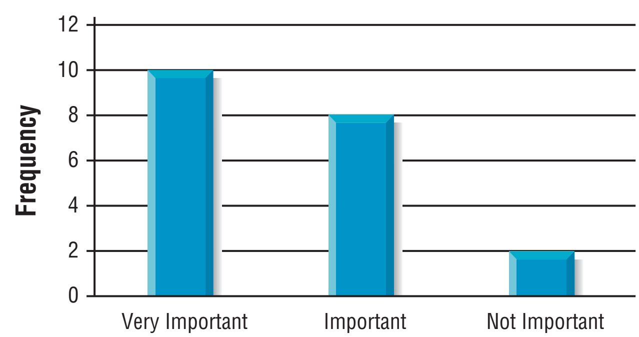


Figure 3 How Important Is Gun Control to You? (*N* = 20)

Bar chart Often used with nominal and ordinal variables, a series of bars represent the different attributes of a variable. The height of each bar reflects either frequencies or percentages for each attribute.

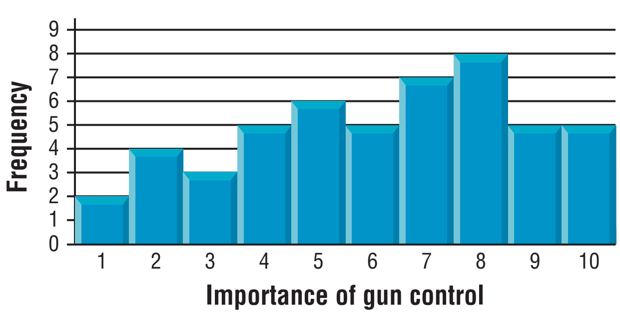
Histogram Used with interval/ratio variables to represent the frequency of each attribute for a variable. Similar to a bar chart.

Bar Charts. The chart in Figure 3 is called a bar chart. Bar charts are often used for ordinal variables because they show trends in a variable. As this chart shows, respondents are more likely to feel that gun control is important or very important than not important.

Histograms. Finally, the chart in Figure 4 is called a histogram. Histograms are generally used to represent interval/ratio data. The height of each bar in the histogram is proportionate to the number of respondents for each response. Unlike bar charts, histograms show all of the data for all responses without groupings and are used to show patterns for interval/ratio variables.

Just as some types of charts are more or less appropriate for visual representations of data, the same is true for statistics. Remember that statistics are nothing more than numeric representations of data that allow us to describe, summarize, and communicate information. Therefore, data collected at different levels of measurement (nominal, ordinal, interval/ratio) are appropriate for different types of statistics. As you may have already guessed, the statistics that can be used to describe a variable depend on whether the variable is operationalized at the nominal, ordinal, or interval/ratio level.

Figure 4 On a Scale of 1 to 10, How Important Is Gun Control to You? (N = 50)



Validity and Reliability

An old saying about statistics goes "garbage in, garbage out." There is a lot of truth to this statement and it raises the issue that sound methodological decision making is the foundation for meaningful statistical analysis. In other words, if we are not careful in how we collect our data and measure our concepts, how can we have any faith in the statistics that come out of that data? Two terms are of particular importance: validity and reliability.

Validity The degree to which a variable measures what we think it is measuring.

Reliability The degree to which a measure yields consistent results.

Validity refers to whether we are measuring what we think we are measuring. Does a variable actually reflect the concept that we think it is?

Reliability refers to the likelihood that a particular measure yields consistent results over time. Is a questionnaire item worded in such a way that respondents interpret it in similar ways?

Suppose a researcher wants to develop a questionnaire item to measure the class standing of a group of college students. When discussing class standing, most people probably think of an individual's relative position in the social structure on the basis of wealth, income, or power. Although *class standing* is not a difficult concept to grasp, it is a difficult concept to operationalize. Here are a couple questions that the researcher might ask:

What was your income last year (in dollars)?

What was your income last week (in dollars)?

Each of these questions does a pretty good job of measuring income, but problems arise with each of them. Are students who made higher amounts of income last year really of higher class standing than those who made lower amounts? Maybe; but it is also possible that some students might not need any income at all because they have a lot of money in the bank. Or perhaps they have very low, or no, income because they receive a check from their parents every other week. In this sense the questions are not valid because there is a good chance that we are not measuring what we think we are measuring.

Now consider the following questions:

Would you say that you live in a working-class, middle-class, or upper-class neighborhood?

Are you financially secure?

Each of these questions does a pretty good job of measuring respondents' subjective class identification, but they, too, have problems. For example, the first question forces respondents to choose from one of three attributes. Often people are reluctant to admit, or do not know, that they are more upper class than middle class. Americans like to fit in. Therefore, it is likely that a vast majority of respondents will answer middle class. If we were to add one more attribute to the number of choices that respondents have, we might find that patterns emerge in the data that cannot be detected with only three attributes. Greater levels of precision, however, do not necessarily increase reliability.

The second question is not reliable because each respondent may read it in a completely different way. For example, respondent A might read the question after having just found out that she needs to buy $700 in books for the upcoming semester. Feelings of financial security may have disappeared in the course of an afternoon upon hearing this news. In this sense, the measure could yield different results depending on the day-to-day finances of respondents instead of on their overall long-range financial position.

When trying to measure difficult concepts like class standing, it is often best to use a variety of measures that can be analyzed independently or combined into a single overall (or composite) measure. For example, many sociologists use a combination of income, education, and occupation to determine respondents' overall socioeconomic status. It is important to remember that measures can be valid without being reliable; and, likewise, measures can be reliable without being valid. The goal is to strive for high levels of both validity and reliability to avoid the problem of "garbage in, garbage out."

Eye on the Applied

Taking the Trash out of Standardized Testing

As is the case of all scientific inquiry, good social science research is dependent on high quality data. This means that data should be both valid and reliable. If a measure of a concept is valid, we are really measuring what we believe to be measuring. Often this is not the case and the consequences of invalid data can be summarized as "garbage in, garbage out."

The No Child Left Behind Act gained widespread support across the nation and has the goal of ensuring that elementary, middle, and high school students achieve minimum levels of education as set by federal and state governments. Standardized tests are used to assess students' abilities. The percent of students who pass these tests is then used as an indicator of schools' ability to educate their students. In this manner, standardized tests provide data that are used to assess both individual students and entire schools, or even school districts.

Standardized testing is believed to assess the cognitive abilities of students and, therefore, act as an indicator of how good a job schools are doing; however, it may not. For example, in many urban high schools, English is not the first language for many students. This can generate lower test scores for non-English-speaking students. Suppose a history course focuses on the transition from agricultural to industrial economies. The students understand the ideas and can write about them, but on the standardized test, the term "agrarian" is more likely to be misinterpreted by students for whom English is not their first language. In this way, the standardized test is biased and not valid because it measures English proficiency, not knowledge of history. It is therefore debatable whether the schools are failing students or whether the standardized tests are failing the schools.

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A second and more likely problem with standardized testing is the assumption that schools with higher percentages of students who pass the exams are better schools. Is a high school in which 87% of the students pass a standardized test a better school than one in which 77% of the students pass? Are the students really learning more? Perhaps they are just learning different content, some of which is assessed on standardized tests and some of which is not.

One trend that most educators agree on is that if you teach only the material that will appear on a test, more of the students will pass that test. Is the school in which 87% of the students pass teaching only the material that will be on the test? If so, then the outcome of standardized testing is to reduce overall learning to a narrowly defined curriculum of items that appear on tests.

In the world of education reform and standardized testing, it is useful not only to understand how to interpret statistics but also to understand the social conditions (political and otherwise) in which those statistics were generated.

Individual Data

Every year, the National Opinion Research Center (NORC) at the University of Chicago conducts its annual General Social Survey (GSS). NORC has conducted this survey every year since 1972 and many of the questionnaire items have remained unchanged. This allows researchers to conduct *longitudinal* analyses (comparing changes in responses over time).

In the 2006 GSS, NORC sampled 4,510 individuals. Not every respondent was asked every question. This means that responses from as few as 2,000 people are used to generalize to the entire population of the United States. Incredible as this may seem, the consistency of these statistics over time indicate that the data gathered is both valid and reliable. Needless to say, NORC researchers must be very diligent in their survey research methodology to ensure that the data are of high quality.

The GSS survey data is known as individual-level data because all of the variables represent characteristics of the individuals that were sampled (sex, race, income level, political party affiliation, occupational prestige score, and their attitudes toward hundreds of different social issues).

Below are three examples of some of the GSS variables and how they are operationalized. The data in these tables represent all the responses from 1972 to 2006. As you can see in Table 11, the variable sex is operationalized as 1 = Male and 2 = Female. The numeric codes (1 and 2) are used in many statistical software programs and allow data entry to be done by entering numeric codes rather than typing out "Male" and "Female." You can see that since 1972, this questionnaire item has been administered to 51,020 respondents and 22,439 of these respondents, 44.0%, have indicated that they are male. Another 28,581 have indicated that they are female.

Tables 11, 12, and 13 are known as frequency tables. Table 12 shows data on respondents' highest degree, an ordinal variable, and Table 13 shows data on respondents' years of education, an interval/ratio variable. As Table 13 shows, interval/ratio variables do not have labels for their codes because the attributes are already in numeric form.

TABLE 11 *Example of a Nominal Variable from the GSS*

SEX — **RESPONDENT'S SEX**

Text of this Question or Item

23. Code respondent's sex

% Valid	% All	N	Value	Label
44.0	44.0	22,439	**1**	MALE
56.0	56.0	28,581	**2**	FEMALE
100.0	100.0	51,020		Total

Properties

Data type: Numeric

Missing-data code: 0

Record/column: 1/108

TABLE 12 *Example of an Ordinal Variable from the GSS*

DEGREE		RS HIGHEST DEGREE		
% Valid	**% All**	**N**	**Value**	**Label**
23.2	23.1	11,777	**0**	LT HIGH SCHOOL
51.7	51.6	26,307	**1**	HIGH SCHOOL
5.1	5.1	2,601	**2**	JUNIOR COLLEGE
13.6	13.6	6,918	**3**	BACHELOR
6.4	6.4	3,253	**4**	GRADUATE
	0.1	29	**8**	DK
	0.3	135	**9**	NA
100.0	100.0	51,020		Total
Properties				
Data type:	Numeric			
Missing-data codes:	7, 8, 9			
Record/column:	1/104			

TABLE 13 *Example of an Interval/Ratio Variable from the GSS*

EDUCATION		HIGHEST YEAR OF SCHOOL COMPLETED		
% Valid	**% All**	**N**	**Value**	**Label**
0.3	0.3	137	**0**	
0.1	0.1	38	**1**	
0.3	0.3	132	**2**	
0.4	0.4	223	**3**	
0.6	0.5	280	**4**	
0.7	0.7	371	**5**	
1.3	1.3	666	**6**	
1.6	1.6	811	**7**	
4.9	4.8	2,470	**8**	
3.5	3.5	1,765	**9**	
4.8	4.8	2,451	**10**	
6.0	6.0	3,065	**11**	
31.1	31.0	15,814	**12**	
8.3	8.3	4,235	**13**	
10.7	10.6	5,422	**14**	
4.3	4.3	2,211	**15**	
11.8	11.8	6,025	**16**	

EDUCATION		HIGHEST YEAR OF SCHOOL COMPLETED		
% Valid	% All	N	Value	Label
2.9	2.9	1,482	**17**	
3.3	3.3	1,660	**18**	
1.3	1.3	648	**19**	
1.9	1.9	962	**20**	
	0.1	64	**98**	DK
	0.2	88	**99**	NA
100.0	100.0	51,020		Total

Properties

Data type: Numeric

Missing-data codes: 97, 98, 99

Record/columns: 1/96-97

Ecological Data

The U.S. Bureau of the Census conducts a national census every 10 years. Census questionnaires come in both long and short form, with most people answering on the short form. Although data are collected from individuals, no data in the census can be analyzed at the level of the individual. Instead, data are tallied by region of the country, state, county, minor civil division, census tracts, block groups, blocks, and other levels of analysis. In this way, census data allow researchers to compare characteristics of states, counties, cities, and so on, without having any knowledge on specific individuals who provided the data. In fact, to protect the identity of people, many census variables are not available at the census tract, block group, or block level.

For example, suppose that you and your family immigrate to the United States from the Dominican Republic and indicate on the census form that you are Dominican. It is likely, in most communities, that you are one of very few Dominican families. Therefore, if data on ethnicity were made available at the block or block group level, other people, using census data, would be able to learn more about your income level, your family makeup, and a host of other data that should be kept private. In other words, if data are made available at too "local" a level, you could be "outed" without your approval. It is therefore important that census data be made available in ways that maintain the anonymity of those who provided it.

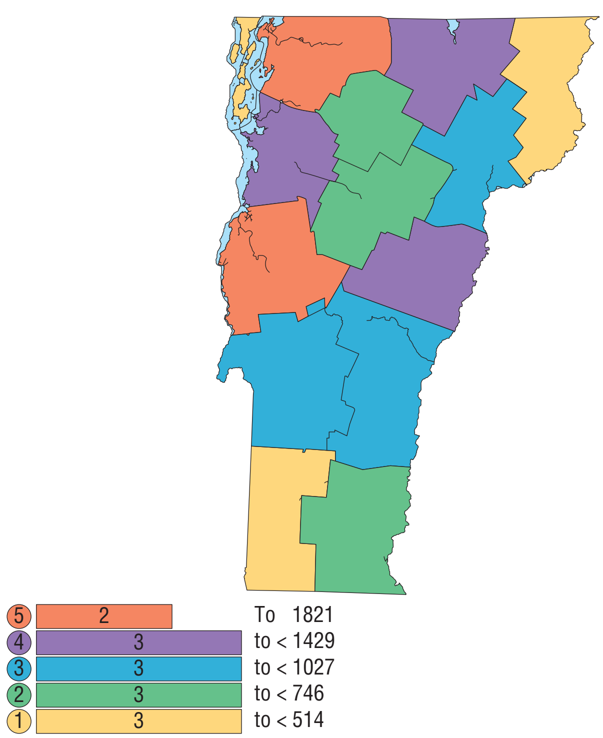
Census data are collected and put into databases in interval/ratio form. For example, suppose we are interested in studying population in the state of Vermont. Communities could be defined as counties, towns and cities, zip codes, or block groups. Suppose we decide to analyze population by county and subsequently divide the state into its 14 counties. The table below shows each of the 13 counties, their total population, and their farm population. We can use these data to calculate the percent of each county's population that lives on farms without revealing who those individuals are.

As Table 14 shows, Vermont's two biggest farm populations are located in Addison and Franklin counties. We do not know, however, where these counties are located or where they lie in relation to other counties with significant farm populations.

TABLE 14 ***Demographic Characteristics of Vermont Counties (N = 14)***

COUNTY	POPULATION	FARM POPULATION	PERCENT FARM POPULATION
Addison	35,974	1,429	4.0
Bennington	36,994	327	.9
Caledonia	29,702	824	2.8
Chittenden	146,571	1,027	.7
Essex	6,459	118	1.8
Franklin	45,417	1,821	4.0
Grand Isle	6,901	200	2.9
Lamoille	23,233	514	2.2
Orange	28,266	1,136	4.0
Orleans	26,277	1,031	3.9
Rutland	63,400	807	1.3
Washington	58,039	688	1.2
Windham	44,216	534	1.2
Windsor	57,418	746	1.3
Total	608,827	11,202	

Figure 5 **Vermont Farm Population by County, 2000**



It is possible, with a variety of different software packages, to map census data to create visual representations that allow researchers to see geographic patterns in data that might not otherwise be noticed. For example, the map in Figure 5 shows that Vermont's farm population is heavily concentrated in two counties, and you can see that these two counties are located in the western part of the state in what is called the Champlain Valley.

When data are represented in both table and graphic formats, we are able to understand it more easily and use it more effectively.

Statistical Uses and Misuses

The Individualistic Fallacy

When working with data representing individuals, it is important not to fall victim to the *individualistic fallacy*. This is a mistake that occurs when researchers infer characteristics of members of a group based on information they obtained from one person in that group. It can be thought of as overgeneralizing.

Consider this example: You go to visit a friend in Boston over spring break. While staying at this friend's house, you overhear a number of conversations and complaints regarding problems with high taxes. When you get home you mention to some other friends that Bostonians are in support of lower taxes. The problem is that you have taken a small (unscientific) sample of Bostonians who are all part of the same group of acquaintances and used their opinions to generalize the whole city of Boston.

The individualistic fallacy can be thought of as generalizing from the individual (or small group) to larger groups.

The Ecological Fallacy

Another term to be considered is called the *ecological fallacy*. The ecological fallacy is a type of error in which characteristics of an area or a region are believed to be the characteristics of the people in that region. An example is as follows: *The state of Oklahoma has one of the largest Native American populations (as a percent of the total population) of all the states. It is also a state that ranks very high in per capita alcohol consumption. Therefore, Native Americans are heavy drinkers.*

It is important, however, to consider other possibilities. Oklahoma is home to a significant number of Indian reservations. Often, these reservations are home to casinos where a significant amount of alcohol consumption takes place by those that gamble. Therefore, it may not be the Native Americans who are consuming all the alcohol. As you can see, the ecological fallacy has the potential to foster and perpetuate stereotypes.

In sum, the ecological fallacy is a mistake that is made when we impose characteristics of a group upon individuals in that group.

EXAMPLE 5 (INDIVIDUAL-LEVEL DATA):

Operationalizing Studiousness

We begin with the concept of *studiousness*. Studious can refer to many different ideas: attending class regularly, studying outside of class time, or doing all of the assigned readings. Suppose that we define it as the number of hours a person spends studying outside of class each week. We can measure this concept on a questionnaire in many different ways.

Consider the following:

Nominal: Do you study in the library? Yes No
Ordinal: How often do you study in the library? Never Sometimes Often
Interval/Ratio: How many hours do you spend studying in the library each week? ______

Individualistic fallacy A type of error that occurs when the characteristics of an individual are imposed upon all of the members of a group to which that individual belongs.

Ecological fallacy A type of error that results from drawing conclusions about individuals from characteristics of a group.

Our conceptual definition of studiousness is the effort that one puts into being a student. Our operational definition of studiousness is done in three different ways. Each has its own advantages and limitations. It is always possible that our operational definitions have little to do with the actual concept with which we began. When our operational definition fails to measure the concept with which we began, we say the variable is lacking *validity*.

For example, if we asked our respondents how many hours they went to the library each week, we could have an invalid measure. It is possible that some students go to the library only to take naps. In this case, a significant number of hours in the library would not indicate greater levels of studiousness. Or it may be that some students indicate that they never go to the library. Does this mean they are not studious? It may be that they have a home office or other place where they do their studying.

If this is the case, it means that our measure only measures part of the concept we intend it to measure. It may be that a student does not use the library at school because they have a better place to study. Or it may mean that they study in other places because they are not afforded the opportunity to study in the library. In either case, we cannot safely conclude that a student who does not study in the library very often is not studious. In this sense, our indicator lacks what is called *content validity* in that it only measures part of what we want it to measure.

Ultimately, social scientists can measure anything, but only with varying degrees of validity and reliability. *Validity* refers to the question "Are we measuring what we think we are measuring?" *Reliability* refers to the question "Can we expect consistent results if our measure is repeated?" In other words, do all our respondents understand the question the same way?

Validity refers to the ability of a measure to accurately reflect the concept it is intended to measure. Reliability refers to the ability of a measure to collect similar data each time it is applied. The following are examples of variables operationalized at different levels.

1. The number of hours you spend studying each day (0, 1, 2, 3, 4, 5, . . .). Ratio
2. How often you would say you study (none, some, a lot). Ordinal
3. Your favorite subject in school. Nominal
4. The frequency of skipping classes (never, sometimes, often). Ordinal
5. The number of classes you skipped last semester (0, 1, 2, 3, 4, ...) Ratio
6. The number of study group meetings you had last semester (0–1, 2–3, 4–5, 6+)

Don't be fooled by #6. Grouped data take the form of ordinal variables.

EXAMPLE 6 (ECOLOGICAL DATA):

Toxic Hazards and Other Locally Unwanted Land Uses

Locally unwanted land uses (or LULUs, as they are often referred to) can take many forms, ranging from toxic waste sites to wind farms. Toxic waste sites in particular pose a wide range of threats to both environmental and human health and include, among others, Superfund sites, state-level hazardous sites, industrial emissions, landfills, incinerators, waste-to-energy plants, tire piles, and trash transfer stations.

A colleague of mine, Daniel Faber from Northeastern University in Boston, and I spent a number of years developing variables to assess the relative contamination of one community to the next. We found that by using only one or two indicators of ecological hazards, say state-level hazardous sites and landfills, our indicator suffered from a lack of content validity. In other words, we were only measuring a portion of the total hazards and could not compare the relative contamination levels in one town to the next.

Ultimately we came up with a composite measure that included 17 different types of ecological hazards. Each type of hazard was awarded a certain number of "points" and when the points were tallied for each community, they represented what we call each community's Ecological Hazard Points, or EHPs. EHP is a ratio variable that in 2005 ranged from 0 in some small remote towns in Massachusetts to about 3,500 in the city of Boston.

Because each community is a different size, we then divided the number of EHPs by the number of square miles of land in each community to get EHP/sq. mile. This allowed us to compare the relative risks that each community faced.

After developing this variable, we were able to test whether lower-income communities and those with significant nonwhite populations were most likely to have high ecological hazard scores. Not surprisingly, they were. To arrive at this conclusion, we divided all the communities into four groups based on the percentage of the total population that was nonwhite. In other words, we began with a ratio variable and recoded it into an ordinal variable.

1. Percent of population that is nonwhite (0, 1, 2, 3, 4 . . . 100) Ratio
2. Percent of population that is nonwhite (0–4.9%, 5–14.9%, 15–24.9%, 25% and greater) Ordinal
3. Number of EHPs/sq. mile in community (0, 1, 2, 3, 4 . . . 127) Ratio
4. Number of EHPs/sq. mile in community (0–5, 6–10, 11– 15, 16 or more) Ordinal

The chart in Figure 6 shows the relationship between Environmental Hazard Points per square mile with the percentage of nonwhite population. In this chart, EHP is operationalized as an interval/ratio variable and race is operationalized as an ordinal variable. As you can see, the number of environmental hazard points increases as the percentage of the population that is nonwhite increases.

The chart in Figure 7 shows the relationship between EHPs *per square mile* with the percentage of nonwhite population (variables operationalized at the ratio level). A "best fit line" has been added to the chart to more clearly show the trend in the data. As in the previous chart, the trend shows a positive association between the two variables—as the percentage of the nonwhite population increases, so does the intensity of ecological hazards.

Sam Kittner/ZUMA Press/Newscom

Figure 6 Environmental Hazard Points by Race

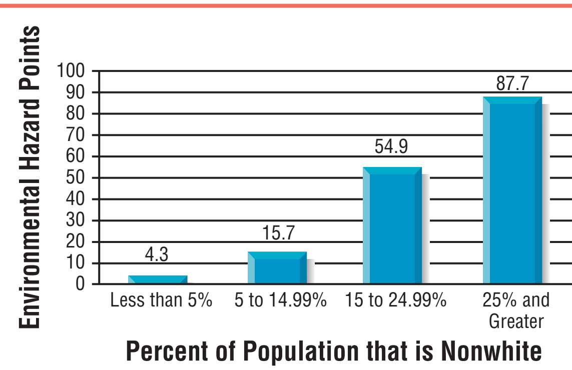
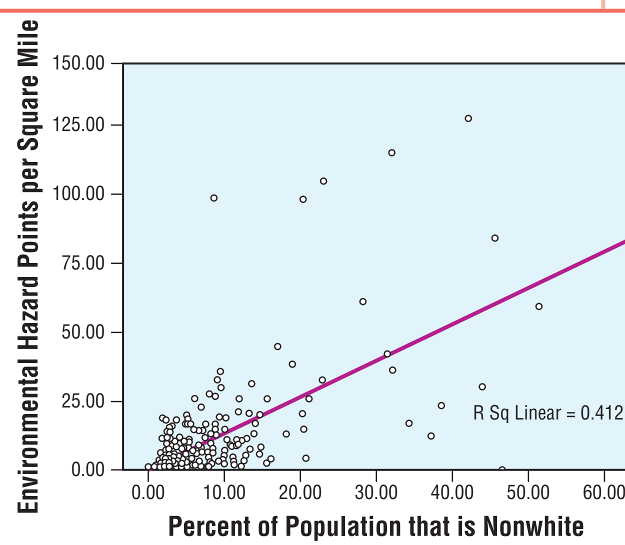


Figure 7 Environmental Hazard Points Per Square Mile by Race



Chapter Summary

This chapter focuses on three main ideas: concepts, variables, and measurement. Concepts are ideas that we have about the world around us; variables are characteristics that vary from case to case; and measurement refers to the ways that we choose to assess the characteristics that vary from case to case. Because concepts are ideas based on what we know, we should expect them to change over time. As Thomas Kuhn argues in his book, *The Structure of Scientific Revolutions*, ideas about the world often change rapidly. Consequently, the ways that we conceptualize and operationalize variables also change over time. Measurement can be done at the nominal, ordinal, and interval/ratio levels. In all three types, the attributes of variables must be collectively exhaustive and mutually exclusive. Two types of data that social scientists work with are individual data and ecological data. While individual data describes characteristics of individuals, ecological data describes characteristics of groups or regions.

Exercises

1. Suppose we conceptualize academic achievement to mean how well a person does in school. Operationalize this concept in two ways.
2. Suppose you want to operationalize racism at the individual level (the degree of racist opinions a person holds). Operationalize this concept in two ways.
3. Come up with a way to operationalize proficiency in Spanish.
4. In your own words, explain what is meant by the term *mutually exclusive*.
5. In your own words, explain what is meant by the term *collectively exhaustive*.
6. Explain the difference between conceptualization and operationalization.
7. What is the problem with the variable below?

THE NUMBER OF BOOKS YOU READ LAST SUMMER
0–2 books
2–4 books
4–6 books
6 or more books

8. What is the problem with the variable below?

THE NUMBER OF BOOKS YOU READ LAST SUMMER
1–2 books
3–5 books
6 or more books

At what level are each of the following variables operationalized?

9. ***Age*** (1, 2, 3 . . .)
10. ***Sex*** (male, female, other)
11. ***Class standing*** (freshman, sophomore, junior, senior, other)
12. ***Marital status*** (married, single, divorced, widowed, other)
13. ***Median Household Income*** (annual household income in dollars)
14. How much you like the food on campus (a lot, some, a little, other)
15. The number of toxic waste sites in your community (0, 1, 2, 3, . . .)
16. The number of toxic waste sites in your community (0, 1–5, 6–10, 11+)
17. ***Your GPA*** (below average, average, above average)
18. Explain why it is best to collect data at the highest level of measurement possible.

Computer Exercises

Go to the National Opinion Research Center's 2006 General Social Survey Codebook at http://sda.berkeley.edu/D3/GSS08/Doc/gs06.htm. Use the information provided in the sequential list of variables to determine whether they are operationalized at the nominal, ordinal, or interval/ratio level.

Variable Name

19. ABANY
20. ACQGAY
21. ACTLIT
22. ADULTS
23. ADVFAM
24. AGE
25. BIBLE
26. BLKJOBS
27. CARSGEN
28. CONCONG
29. MUSICALS
30. MYFAITH
31. NATBORN
32. NUMDAYS
33. PAEDUC
34. PAIDSEX
35. POLVIEWS
36. RACE
37. SEX
38. TRUST

Now You Try It Answer

1: All heavenly bodies rise in the east and fall in the west. This is a function of the rotation of the Earth on its axes. If you have a globe and a flashlight, you can shine the light on the globe, imagining the light is sunlight, and spin the globe to better understand this.

Key Terms

Alienation
Anomalies
Anomie
Attributes
Bar chart
Collectively exhaustive
Concept
Conceptualization
Dependent variable
Direction of effect
Ecological fallacy
Histogram
Hypothesis
Independent variable
Individualistic fallacy
Interval/ratio
Levels of measurement
Life chances
Mutually exclusive
Nominal
Operationalization
Ordinal
Pie chart
Reliability
Statistics
Units of analysis
Validity
Variable

Works Cited

Best, J. (2001). *Damned Lies and Statistics: Untangling Numbers from the Media, Politicians, and Activists.* Berkeley: University of California Press.

Bullard, R. D. (1994). *Unequal Protection: Environmental Justice and Communities of Color.* San Francisco: Sierra Club Books.

Centers for Diseases Control and Prevention. (2009, May 15). *FastStats.* Atlanta, GA: Autho. Retrieved September 18, 2008, from http://www.cdc.gov/nchs/faststats/suicide.htm

Durkheim, E. (1951). *Suicide: A Study in Sociology* (J. A. Spaulding & G. Simpson, Trans.; G. Simpson, Ed.). New York: Free Press. (Original work published 1897)

Faber, D. R., & Krieg, E. J. (2005). *Unequal Exposure to Ecological Hazards 2005: Environmental Injustices in the Commonwealth of Massachusetts.* Boston: Philanthropy and Environmental Justice Research Project, Northeastern University.

Kimmel, M. (2008). *Guyland: The Perilous World Where Boys Become Men.* New York: HarperCollins.

Kuhn, T. (1962). *The Structure of Scientific Revolutions.* Chicago: University of Chicago Press.

Loewen, J. (2005). *Sundown Towns: A Hidden Dimension of American Racism.* New York: New Press.

Marx, K. (1998). *Capital* (D. McLelland, Ed.). New York: Oxford University Press.

Newton, I. S. (1848). *Principia.* New York: D. Adee.

Weber, M. (2004). *The Protestant Ethic and the Spirit of Capitalism.* New York: Routledge.